

## Born approximation

The Born approximation gives scattering amplitude  $\phi(\mathbf{p})$  as

$$\phi(\mathbf{p}) = -\frac{m}{2\pi\hbar^2} \int \exp\left(\frac{i\mathbf{p} \cdot \mathbf{y}}{\hbar}\right) V(\mathbf{y}) d\mathbf{y} \quad (1)$$

For Coulomb potential

$$V(\mathbf{y}) = -\frac{Z\alpha\hbar c}{|\mathbf{y}|}$$

the solution is

$$\phi(\mathbf{p}) = -\frac{2mZ\alpha\hbar c}{|\mathbf{p}|^2}$$

By the identity

$$|\mathbf{p}|^2 = 4mE(1 - \cos \theta)$$

we have

$$\phi(\theta) = -\frac{Z\alpha\hbar c}{2E(1 - \cos \theta)}$$

Hence the Rutherford cross section

$$\frac{d\sigma}{d\Omega} = |\phi(\theta)|^2 = \left[ \frac{Z\alpha\hbar c}{2E(1 - \cos \theta)} \right]^2 \quad (2)$$

Let us derive (1) from the time-independent Schrodinger equation

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{x}) + V(\mathbf{x})\psi(\mathbf{x}) = E\psi(\mathbf{x})$$

Rewrite the Schrodinger equation as

$$(\nabla^2 + k^2) \psi(\mathbf{x}) = \frac{2m}{\hbar^2} V(\mathbf{x})\psi(\mathbf{x})$$

where

$$k = \frac{\sqrt{2mE}}{\hbar}$$

Now we are going to make the Schrodinger equation even more complicated.

By the sifting property of  $\delta(\mathbf{x} - \mathbf{y})$  we can write

$$(\nabla^2 + k^2) \psi(\mathbf{x}) = \frac{2m}{\hbar^2} \int \delta(\mathbf{x} - \mathbf{y}) V(\mathbf{y})\psi(\mathbf{y}) d\mathbf{y}$$

Let  $G(\mathbf{x})$  be a Green's function such that

$$(\nabla^2 + k^2) G(\mathbf{x}) = \delta(\mathbf{x})$$

Then by substitution

$$(\nabla^2 + k^2) \psi(\mathbf{x}) = \frac{2m}{\hbar^2} \int (\nabla^2 + k^2) G(\mathbf{x} - \mathbf{y}) V(\mathbf{y})\psi(\mathbf{y}) d\mathbf{y}$$

By linearity

$$(\nabla^2 + k^2) \psi(\mathbf{x}) = (\nabla^2 + k^2) \frac{2m}{\hbar^2} \int G(\mathbf{x} - \mathbf{y}) V(\mathbf{y}) \psi(\mathbf{y}) d\mathbf{y}$$

Hence

$$\psi(\mathbf{x}) = \frac{2m}{\hbar^2} \int G(\mathbf{x} - \mathbf{y}) V(\mathbf{y}) \psi(\mathbf{y}) d\mathbf{y}$$

The well-known solution for  $G(\mathbf{x})$  is

$$G(\mathbf{x}) = -\frac{\exp(ik|\mathbf{x}|)}{4\pi|\mathbf{x}|}$$

Hence

$$\psi(\mathbf{x}) = -\frac{m}{2\pi\hbar^2} \int \frac{\exp(ik|\mathbf{x} - \mathbf{y}|)}{|\mathbf{x} - \mathbf{y}|} V(\mathbf{y}) \psi(\mathbf{y}) d\mathbf{y}$$

For potentials  $V(\mathbf{y}) \approx 0$  for large  $|\mathbf{y}|$  we have the approximation

$$\frac{\exp(ik|\mathbf{x} - \mathbf{y}|)}{|\mathbf{x} - \mathbf{y}|} V(\mathbf{y}) \approx \frac{\exp(ik|\mathbf{x}|)}{|\mathbf{x}|} \exp\left(-\frac{ik\mathbf{x} \cdot \mathbf{y}}{|\mathbf{x}|}\right) V(\mathbf{y})$$

Hence

$$\psi(\mathbf{x}) = -\frac{m}{2\pi\hbar^2} \frac{\exp(ik|\mathbf{x}|)}{|\mathbf{x}|} \int \exp\left(-\frac{ik\mathbf{x} \cdot \mathbf{y}}{|\mathbf{x}|}\right) V(\mathbf{y}) \psi(\mathbf{y}) d\mathbf{y}$$

For the Born approximation we use

$$\psi(\mathbf{y}) = \exp\left(\frac{ik\mathbf{x}' \cdot \mathbf{y}}{|\mathbf{x}'|}\right)$$

Hence

$$\psi(\mathbf{x}) = -\frac{m}{2\pi\hbar^2} \frac{\exp(ik|\mathbf{x}|)}{|\mathbf{x}|} \int \exp\left(\frac{ik\mathbf{x}' \cdot \mathbf{y}}{|\mathbf{x}'|} - \frac{ik\mathbf{x} \cdot \mathbf{y}}{|\mathbf{x}|}\right) V(\mathbf{y}) d\mathbf{y}$$

Let  $\mathbf{p}$  be momentum transfer

$$\mathbf{p} = \hbar k \left( \frac{\mathbf{x}'}{|\mathbf{x}'|} - \frac{\mathbf{x}}{|\mathbf{x}|} \right)$$

Then by substitution

$$\psi(\mathbf{x}) = -\frac{m}{2\pi\hbar^2} \frac{\exp(ik|\mathbf{x}|)}{|\mathbf{x}|} \int \exp\left(\frac{i\mathbf{p} \cdot \mathbf{y}}{\hbar}\right) V(\mathbf{y}) d\mathbf{y}$$

Partition  $\psi(\mathbf{x})$  as

$$\psi(\mathbf{x}) = \frac{\exp(ik|\mathbf{x}|)}{|\mathbf{x}|} \phi(\mathbf{p})$$

where  $\phi(\mathbf{p})$  is the scattering amplitude

$$\phi(\mathbf{p}) = -\frac{m}{2\pi\hbar^2} \int \exp\left(\frac{i\mathbf{p} \cdot \mathbf{y}}{\hbar}\right) V(\mathbf{y}) d\mathbf{y}$$